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## Thermal Measurements of a Rail-Cask-Size Pipe-Calorimeter in Jet Fuel Fires

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### ABSTRACT

A series of fire tests was conducted involving a 2.44-m-(8-ft)-diameter, 4.57-m-(15-ft)-long, 2.54-cm-(1-inch)-wall thickness pipe calorimeter suspended 1-m above a 7.93-m-diameter pool that contained 7.57 m<sup>3</sup> (2000 gallons) of jet fuel. The wind conditions, calorimeter temperature, participating media temperature and speed, and radiant heat flux, were measured at several locations as functions of time in three fire tests. The first two had relatively light winds and lasted roughly 40 minutes, while the third had much stronger winds and consumed the fuel in 25 minutes. The purpose of this paper is to describe the experimental facilities and certain fire characteristics. The large amount of data acquired cannot be fully presented in this paper. A website is available by contacting the first author so that the full data set may be used to quantitatively benchmark large-fire simulations and models.

### I. INTRODUCTION

Federal regulations (10CFR71) require packages that transport large quantities of radioactive materials (Type B) to withstand a test sequence consisting of a 9-m drop onto an

unyielding surface, a 1-m drop onto a puncture bar, a 30-minute, 800°C fully-engulfing fire and water immersion [US Nuclear Regulatory Commission 2009]. At the conclusion of this sequence, the confinement, shielding and criticality functions of the package must be maintained. This regulatory sequence is estimated to be more severe than 99.4% of all transportation accidents [Fischer et al. 1987]. However, risk assessment studies must consider the likelihood and consequence of all scenarios that are possible during transport campaigns [Sprung et al. 2000].

The thermal performance of packages for regulatory testing and other severe events is evaluated by both testing and analysis [ASTM Standard Practice (E2230-08)]. Analyses typically involve construction of finite element (FE) thermal models of intact or damaged packages [General Atomics 1998]. First the steady state package temperatures are calculated for normal conditions of transport. These temperatures are used as initial conditions for a transient calculation that determines the time dependent package temperatures during a fire. Finally, the package temperatures at the end of the fire are used as initial conditions for a post-fire calculation.

A variety of models have been used to predict the heat transfer from large, long-duration fires to a cask. The Code of Federal Regulations, 10CFR71 [US Nuclear Regulatory Commission 2009], indicates that the cask must be fully engulfed in a fire environment whose average temperature and radiant emissivity must be at least 800°C and 0.9, respectively. However, this fire model may not be useful

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for risk assessment studies that consider a variety of cask locations relative to the fire and different wind conditions.

Sophisticated fire physics computational fluid dynamics (CFD) and radiation heat transfer codes such as ARCHES (developed at the University of Utah C-SAFE program [Spinti et al. 2008]) and general purpose commercial packages such as Fluent have the capability to predict heat transfer to large objects for a variety of wind conditions and package placements. The Container Analysis Fire Environment (CAFE) was developed at Sandia National Laboratories (SNL) specifically to predict the response of casks to a variety of fire conditions for risk studies [Suo-Anttila et al. 2005]. It links a finite element cask model with a CFD/diffuse radiation heat transfer fire model. Computational fire models must be benchmarked against experimental measurements before they can be used with confidence to predict the response of casks or other large objects in severe accident conditions [del Valle et al. 2007].

Kramer et al. [2000] measured the temperature response of a 1.2-m-(4-ft)-diameter, 4.6-m-(15-ft)-long, 2.54-cm-(1-inch)-wall-thickness mild-steel pipe calorimeter suspended 1-m above a 7.2-m-(14-ft)-diameter pool of jet fuel. The calorimeter was roughly the size of the body of a package used to transport spent nuclear fuel by truck. The calorimeter temperature and wind conditions were measured during two 30-minute fires. The calorimeter and fire were encircled by a porous barrier, and the wind conditions were measured outside. The barrier was intended to reduce the effect of wind, but measurements show that it did not eliminate that effect entirely. It is complicated to quantify the effect of wind on the fire because it is difficult to model the wind fence. Blanchet et al. [2006] measured the response of the same pipe calorimeter in a series of well defined fire conditions without a wind barrier.

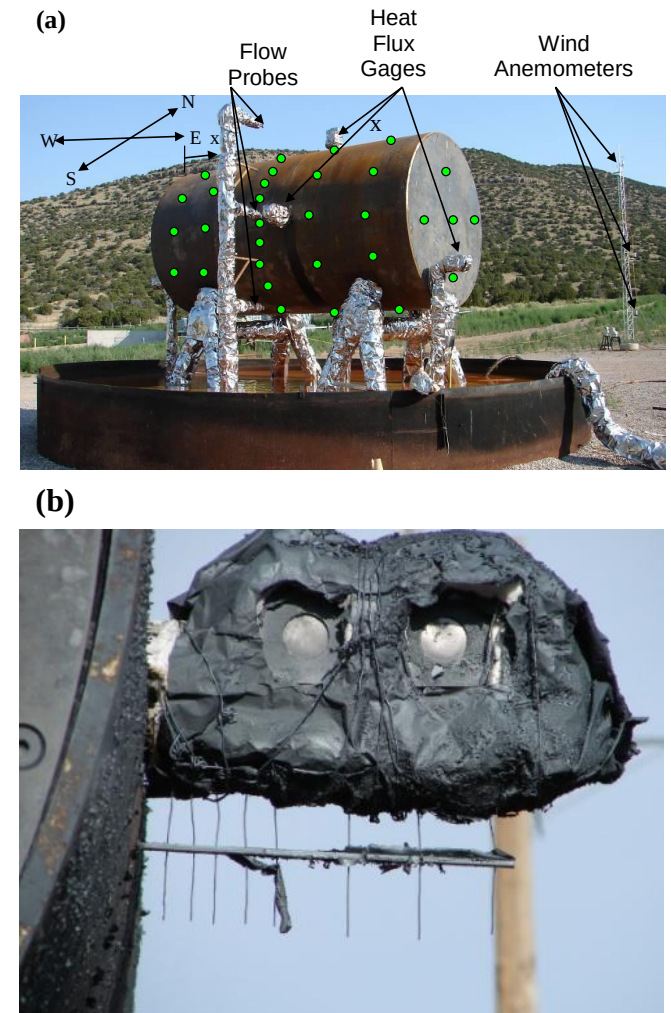
In the summer of 2007, a much larger 2.44-m-(8-ft)-diameter, 4.57-m-(15-ft)-long pipe calorimeter, which is roughly the size of a spent fuel rail transport package body, was suspended over a 7.93-m-(26-ft)-diameter jet fuel pool. The calorimeter interior surface temperature, wind speed and direction, fire gas (participating media) temperature, flow speed, and radiant heat flux, were measured at several locations versus time. These measurements were performed in three different fire tests, and no wind barriers were employed. The tests experienced different wind conditions, and lasted between 25 and 41 minutes. The experiments were performed to acquire data to benchmark large-fire heat transfer simulations and models.

The purpose of this paper is to describe the experimental facilities, the wind conditions, and summarize certain measured characteristics of the fires. The large amount of data acquired in the experiments cannot be fully presented in a single paper. A website containing the full data set is available by contacting the first author. It may be used by other investigators to quantitatively benchmark large-fire simulations and models. The system used in the website for identifying each measurement device is described in the paper.

## II. EXPERIMENTAL FACILITY

Figure 1 shows photographs of the fire test facility. It was located at the Sandia National Laboratories Lurance

Canyon Burn Site. Figure 1a show the 2.44-m-diameter, 4.57-m-long pipe calorimeter supported by an insulated stand above a 7.93-m-diameter, 1-m-tall tank. The largest horizontal distance between the calorimeter and the edge of the tank was 2.7 m on the calorimeter sides, and 1.7 m on the ends. These distances are consistent with the requirements of federal regulations (10CFR71) used to license nuclear material transport packages.



**Figure 1 Photographs of experimental facility. (a) Calorimeter, fuel tank and external instrumentation. (b) Fixture on the north side of the calorimeter with two Sandia Heat Flux Gages (SHFG) and a rake of 8 participating-media thermocouples below.**

The pipe and lid walls were constructed from uncoated 2.54-cm-thick A36 carbon steel (this wall thickness is much less than that of a transport package). The endplates were bolted to brackets welded to the interior of the cylindrical body. The endplate outer surfaces were flush with the cylinder end. Compass directions are also shown in the figure and the calorimeter axis was aligned east-west. The x-axis shown in the figure is also aligned east-west. Its origin  $x = 0$  is at the outer surface of the west (secondary) lid, and  $x$  increases in the eastward direction.

Labels in Fig. 1a identify three flow probes (all supported by a single vertical stand), three heat flux gage fixtures (two of which are attached to the calorimeter, the

other is supported by a stand), and three wind anemometers on a single tower. All instrument and calorimeter support stands that are in the fire are insulated and wrapped with stainless steel foil. Dots on the calorimeter show the approximate locations where thermocouples were strap welded to its interior surface.

Figure 1b shows a Sandia Heat Flux Gage (SHFG) fixture on the north side of the calorimeter (not seen in Fig. 1a) after it was coated with soot during one of the fire tests (Blanchet et al. 2002). The two circles are active heat flux measurement surfaces. Eight participating-media thermocouples hang below that fixture, and they are located 1, 3, 6, 9, 12, 20, 28 and 35 cm from the calorimeter wall.

Four anemometer towers were located directly north, south, east and west of the calorimeter, 24 m (80 ft) from its center (only the north tower is visible in Fig. 1a). Each tower supported three ultrasonic wind anemometers (R.M. Young Company, Model 81000), 2, 5 and 10 m above the ground. Each anemometer gave three components of wind velocity,  $u$  (to the west),  $v$  (to the south) and  $w$  (upward), as well as local temperature. The uncertainty of the velocity measurement is  $\pm 1\%$ .

Three bidirectional flow probes [McCaffrey and Heskestad, 1976; Liu et al., 1990; Sette 2005] were placed in the plane perpendicular to the calorimeter axis at  $x = 2.74$  m (all three are seen in Fig. 1a). They were all located on a vertical line 41 cm away from the south side of the calorimeter. The bottom, middle and top probes were, respectively, 130 m below, 8 cm below, and 114 cm above the calorimeter axis. Each flow probe consisted of a 2.5-mm-long, 1.3-mm-diameter stainless steel tube whose axis was vertical. A wall at the middle of the tube divided it into two cups. The pressure difference between the two cups is measured to evaluate the difference between the stagnation and sub-static pressures,  $\Delta P_{FP}$ . The local temperature is measured to estimate the gas density  $\rho$  using the ideal gas equation of state and an atmospheric pressure of 86 kPa (at 1930 m elevation). The local vertical component of the gas speed is calculated as

$$S_{BDFP} = 1.07 \sqrt{\frac{2\Delta P_{FP}}{\rho}},$$

with an uncertainty of 10%. The gas density is calculated using the ideal gas law and the molar mass of air.

Fifty-eight thermocouples were strap welded to the calorimeter interior surface. All thermocouples used in these tests are Type-k, stainless-steel-sheathed, ungrounded, and insulated using high-purity magnesium oxide. The uncertainty of all thermocouple measurements was the greater of  $\pm 1.1^\circ\text{C}$  or 0.4 percent of the reading in centigrade. There is an additional bias error due to the strap-weld mounting technique.

Forty-eight of the interior surface thermocouples were placed on five rings. Rings 1 to 5 were located at  $x = 419, 324, 229, 133$  and  $38.1$  cm, respectively. All rings had thermocouples at angular positions  $\phi = 0, 45, 90, 135, 180, 225, 270$  and  $315^\circ$ . The location  $\phi = 0$  is at the north side of the calorimeter, and  $\phi$  increases in the clockwise direction as

viewed from the west (for example  $\phi = 90^\circ$  is at the top of the calorimeter). Ring 3 had eight additional thermocouples halfway between each of those angular locations. Each thermocouple is given a four digit identification number. The first digit is the ring number, while the other three indicate the angular position in degrees. For example, thermocouple number TC2090 is the one on Ring 2 at the top of the calorimeter.

The primary (east) and secondary (west) lids had thermocouples at their centers, and four more 88 cm away, directly north, above, south and below the center. On the primary lid these five thermocouples are identified as PLcenter, PL000, PL090, PL180, and PL270, respectively. The secondary lid thermocouples use the same identification system except "SL" is used as the prefix.

Twelve additional thermocouples were embedded in the calorimeter wall near thermocouples 3000, 3090, 3180 and 3270 ( $\phi = 0, 90, 180$  and  $270^\circ$  of Ring 3). At each of these locations, 7.9, 15.9, and 24.6 mm deep holes were drilled into the calorimeter interior surface. The thermocouples were placed at the bottom of these holes and strap welded in place. The last digit of the identification number for the thermocouples 7.9, 15.9, and 24.6 mm beneath the interior surface is 1, 2, and 3, respectively. These thermocouples can be used with an inverse conduction calculation [Blackwell et al. 1987] to obtain better estimates of the calorimeter exterior surface temperature than can be determined from the thermocouples that were strap welded to the interior surface. However data from these thermocouples are not discussed in the current paper.

The thermocouple lead wires were fastened to the calorimeter interior walls. On the calorimeter body they were routed along the five rings and then along the bottom of the calorimeter to a slot in the primary lid. On the two lids the leads ran along horizontal and vertical lines. The thermocouples and lead wire were backed by three 2.5-cm-(1-inch)-thick, 30-cm-wide strips of Kaowool insulation inside the calorimeter to reduce heat transfer with the hollow calorimeter interior. Outside the calorimeter the wire bundle entered the fuel/water pool directly below the exit slot, and exited the pool at its edge. The bundle was insulated and water-cooled outside of the calorimeter.

Sandia Heat Flux Gages (SHFG), Fig. 1b, are used to measure radiant heat flux of the participating media near the calorimeter surfaces [Blanchet et al. 2002]. Each gage is a 10-cm-long, 10-cm-diameter tube. Double sided gages have 0.25-mm thick 304 stainless steel foils on each end, while single sided gages have the foil on only one end. The foils have thermocouples attached to their interior surfaces, and are backed with insulation. The cylinder interior volume is filled with insulation. The foils rapidly approach thermal equilibrium with their surroundings. The thermocouple temperature is an indication of the radiant heat flux incident on the foils. The cylindrical surfaces of the fixtures are covered with fiberglass insulation and stainless steel foil to protect them during the fire test (Fig. 1b).

Eight double sided SHFG's were placed in the plane normal to the calorimeter axis at  $x = 2.74$  m (between thermocouple Rings 2 and 3). They were located at angular



positions  $\phi = 0, 90, 180$  and  $270^\circ$ . At angular positions  $\phi = 0^\circ$  (north side),  $90^\circ$  (top), and  $180^\circ$  (south side), radial poles were attached to the calorimeter and two double-sided gages were connected to each pole. At angular positions  $\phi = 270^\circ$  (bottom), two double sided gages were supported by a stand that stood in the tank beneath the calorimeter. The active foils of all these gages faced east and west, and their centers were 15 and 30 cm away from the calorimeter exterior surface. Four digit identification numbers were given to the heat flux gages. The first digit for the east facing gages was 6, and it was 7 for the west facing gages. The next two digits were the first two digits of the angular position. The last digit was 1 for the gage closer to the calorimeter, and 2 for the one further. For example, gage HFG7092 was the west-facing gage at the top of the calorimeter, whose center was 30 cm away from the calorimeter surface.

Four more single sided SHFGs were placed outside the east (primary) and west (secondary) lids. Their active foils faced north, and they were located 53 cm below and 5 cm south of the lid centers. The identification numbers of the gages whose centers are 15 and 30 cm from the surface are HFG8001 and HFG8002, respectively. The gages closer and further from the secondary lids are HFG9001 and HFG9002.

### III. EXPERIMENTAL PROCEDURE

The Lurance Canyon Burn Site is known to have predominately westward (down canyon) winds at night and eastward (up canyon) winds during the day. The wind conditions are typically very light in the early morning while the wind changes direction. For this reason Tests 1 and 2 were initiated at 7:10 and 7:20 am on July 12 and August 1, 2007 in order for the winds to be light. Test 3 was started at 8:25 am on September 13, 2007 so that stronger winds would be encountered.

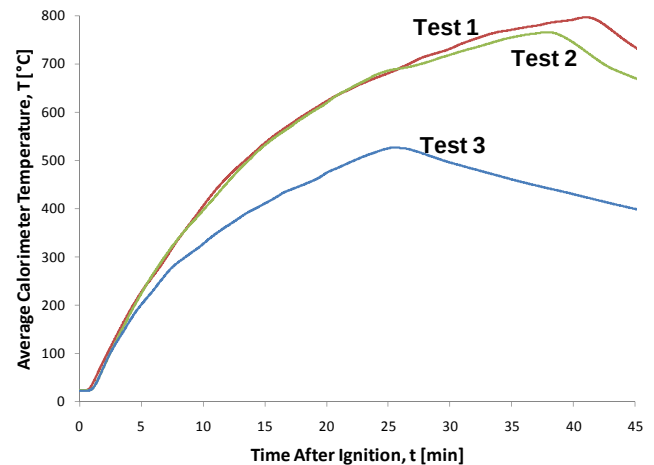
Before each test the tank was filled with water so that its surface was 115.2 cm below the bottom of the calorimeter. A volume of  $7.57 \text{ m}^3$  of jet propulsion (JP-8) fuel was then placed in the tank. It floated on top of the water bringing, the liquid surface 1 m below the calorimeter. This distance is consistent with the requirements of the 10CFR71.73 federal regulations used to certify nuclear waste transport packages.

The data acquisition system recorded all temperature, pressure and wind speed data at one second intervals. The system was activated before the fire was initiated. A propane torch was used to ignite the fuel at the south east region of the pool. Several attempts were required before the flames were self sustaining. It then took roughly one minute for the flames to spread across the pool.

### IV. RESULTS

The temperatures of the thermocouples located outside the calorimeter (the participating-media rake, bi-bidirectional flow probes, and Sandia Heat Flux Gages) were examined to determine when the fire began. The ignition time was defined as the first time when the measured external temperatures exhibited a sustained rise. In all three tests this was one minute after the data acquisition system was activated.

Figure 2 shows the average calorimeter temperature versus time after fire ignition for all three tests. This is the average of the 58 thermocouples on the calorimeter body and lid interior surfaces. The average temperature was between 22 and  $25^\circ\text{C}$  before the tests. It took 0.2 to 0.6 minutes after the fire started before for the average temperature exhibited a measureable rise. Tests 1, 2 and 3 exhibited their maximum average calorimeter temperatures of  $798^\circ\text{C}$ ,  $766^\circ\text{C}$  and  $526^\circ\text{C}$  at  $t = 40.5, 37.6$  and  $25.3$  min, respectively. The exterior instrument temperatures also began to exhibit sustained decreases within 1.6 minutes of these times. This is consistent with the video record that showed the flames took several minutes to extinguish.



**Figure 2 Average calorimeter temperature versus time after fire ignition for Tests 1, 2 and 3.**

For the purposes of our analysis we define the fire duration,  $D$ , as the time after ignition when the average calorimeter temperature exhibited its maximum value. These durations are summarized in one row of Table 1.

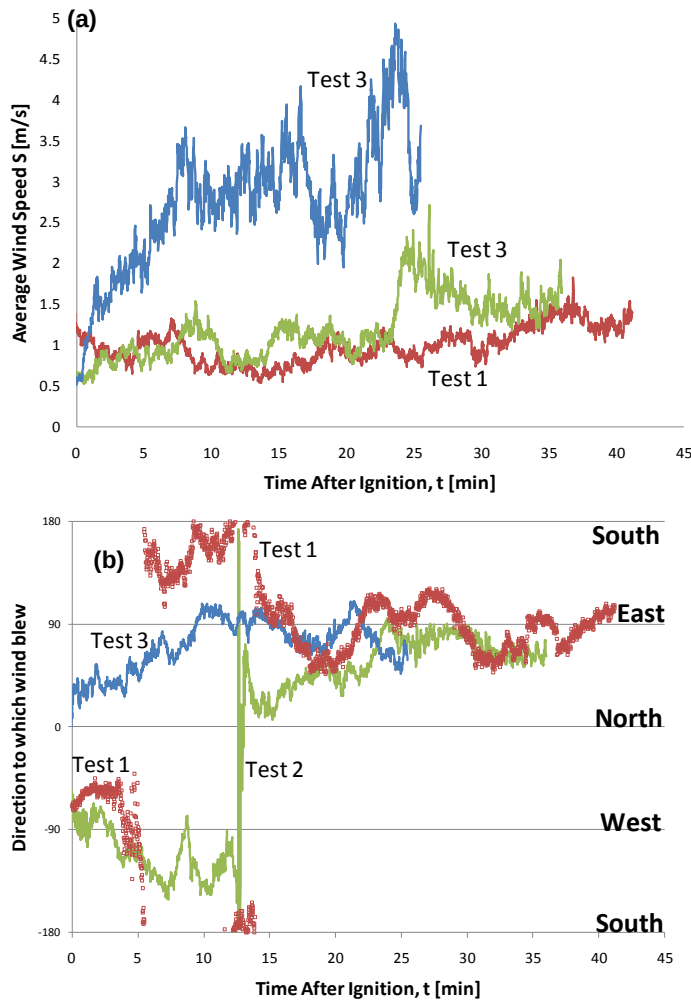
**Table 1 Fire duration, Average calorimeter temperature rise, and Average Wind speed for Tests 1, 2 and 3**

|                                                          | Test |      |      |
|----------------------------------------------------------|------|------|------|
|                                                          | 1    | 2    | 3    |
| Fire Duration, $D$ [min]                                 | 40.5 | 37.6 | 25.3 |
| Average Cal Temp Rise, $\Delta T_A$ [ $^\circ\text{C}$ ] |      |      |      |
| At end of fire ( $t = D$ )                               | 773  | 741  | 503  |
| At $t = 25$ min                                          | 661  | 664  | 502  |
| Average Wind Speed, $S_A$ [m/s]                          | 0.99 | 1.22 | 2.9  |

The average calorimeter temperature rise  $\Delta T_A$  for each test is also summarized in Table 1. It is the average calorimeter temperature at a given time (Fig. 2) minus the average at the beginning of the test. This quantity is an indication of the amount of energy delivered to the calorimeter in that time. Transportation risk assessment models must predict this to accurately calculate the response of a transport package in a fire. Results in Table 1 are given for  $t = D$  (the end of each test) and for  $t = 25$  min (this time was chosen because all three tests lasted at least that long). The fire duration and average calorimeter temperature rise are

significantly smaller for Test 3 than they are for the other tests. To understand these differences we must consider the wind conditions.

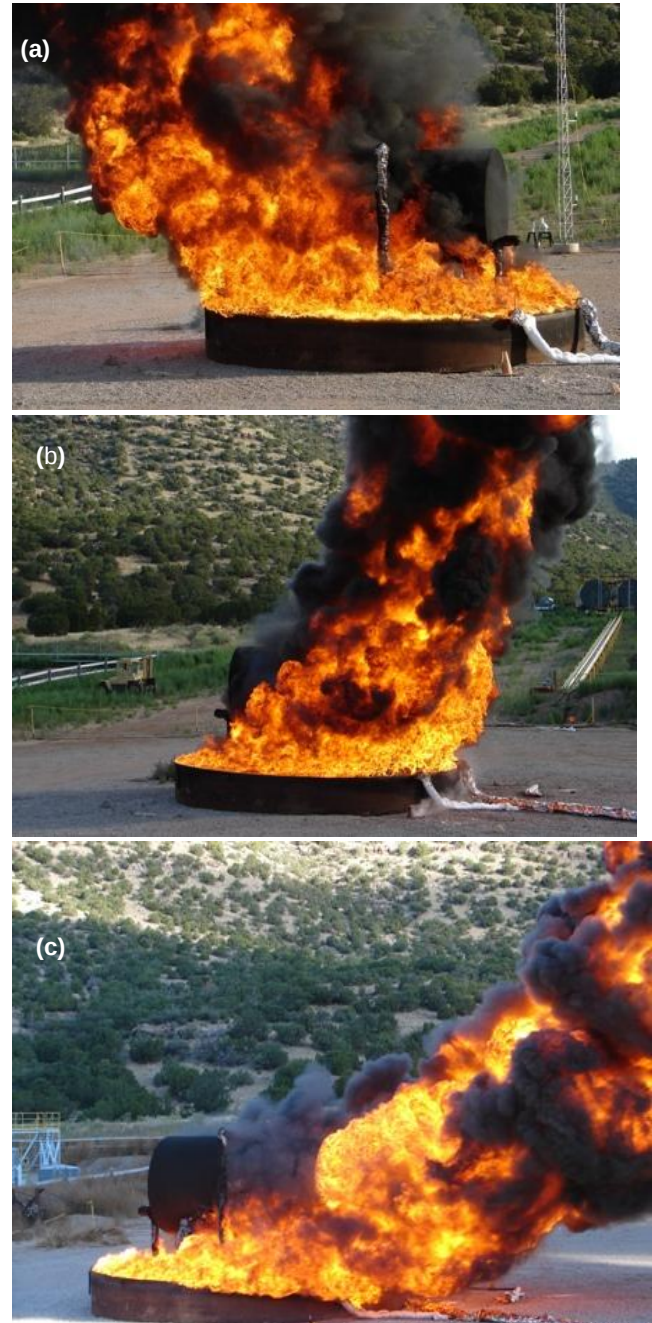
Figures 3a and 3b show the wind speed and compass direction versus time for the three tests. The wind speed at each time was determined by averaging the speed measured by all anemometers at that time. The direction was determined by averaging the  $u$  (westward) and  $v$  (southward) speed components of all the anemometers, and then finding the direction of the resulting vector. In Tests 1 and 2, the anemometer at the top of the North pole exhibited occasional anomalous spikes. In Test 3 the middle and top anemometers on the East pole and the one in the middle of the North pole showed multiple spikes. The data from those anemometers were not used for the tests in which they exhibited spikes.



**Figure 3 Time dependent wind conditions for Tests 1, 2 and 3(a) Speed (b) Direction**

The wind speed was between 0.5 and 1.2 m/s at the beginning of all three tests. The speed during Test 1 remained fairly constant, and the average over the test was 0.99 m/s. The wind speed of Test 2 increased somewhat at  $t = 23$  minutes, and its average was 1.22 m/s. The speed of Test 3 increased dramatically after the fire started and was highly unsteady. Its average was 2.9 m/s. The average speeds  $S_A$  for each test are summarized in Table 1.

For both Test 1 and 2, the wind blew toward the west at the beginning and toward the East at the end. In Test 1 the wind switched directions during the time period  $t = 5$  to 15 minutes. In Test 2 the switch occurred over the period  $t = 13$  to 24 minutes. For Test 3, the wind direction changed from northeast to the east over the period  $t = 0$  to 13 minutes, and blew to the east thereafter.

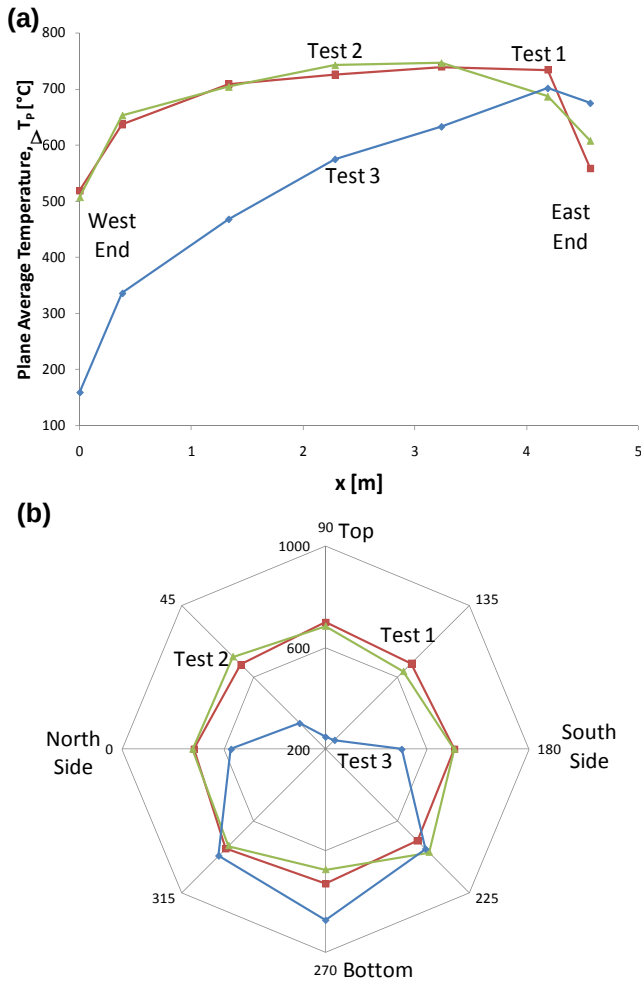


**Figure 4 Fire Photographs (a) Tests 1 and 2, light winds to the west, (b) Tests 1 and 2, light winds to the east, (c) Test 3 stronger wind to the east.**

Figure 4 shows photographs of the fire test taken by an observer located southeast of the calorimeter. Figures 4a and 4b represent the light wind that blew to the west and then the east in Tests 1 and 2. The calorimeter was mostly engulfed during those fires. Figure 4c shows that the stronger

wind conditions that existed in Test 3 tilted the fire so that the calorimeter's west end and top were substantially un-engulfed.

Figure 5 shows regionally-averaged calorimeter temperatures at  $t = 25$  minutes. Figure 5a shows the temperatures measured on the calorimeter lids and rings versus the x-coordinate of these regions. The profiles for Tests 1 and 2 are similar. The temperature of the calorimeter body is fairly uniform at roughly  $700^{\circ}\text{C}$ , but the two lids are cooler. In Test 3 the west (windward) end is less engulfed (Fig. 4c) and substantially cooler than the east (leeward) end. Once again, the lids are cooler than the calorimeter rings near those lids. Finally, the temperature of the east end of the calorimeter body is nearly the same in all three tests. This may be because that end was engulfed in all three fires.



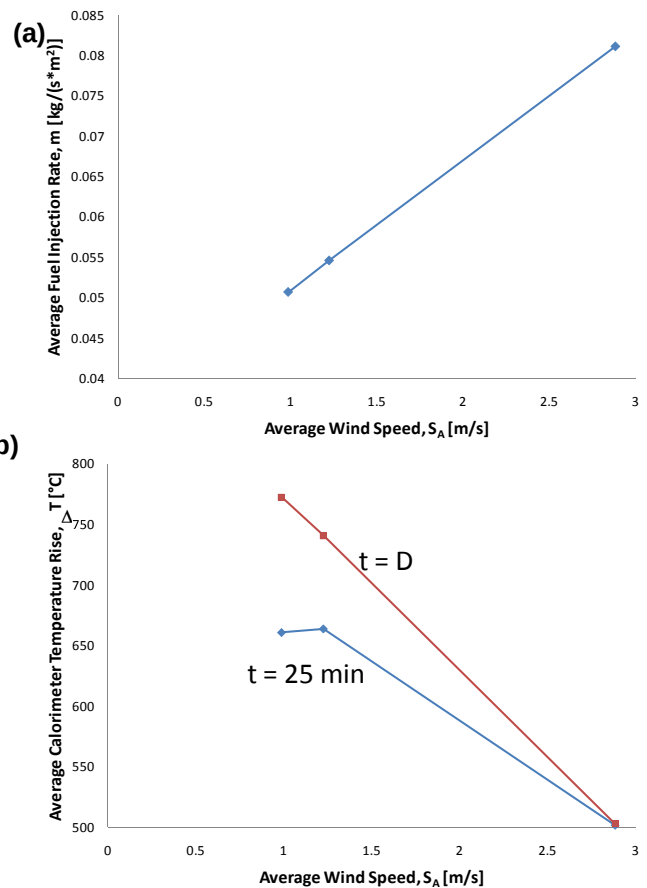
**Figure 5 Regionally-averaged temperature profiles for Tests 1, 2 and 3 at  $t = 25$  min. (a) Average on calorimeter rings and lids versus axial locations, (b) average along lines parallel to calorimeter axis versus angular position**

Figure 5b shows the temperature of the calorimeter body averaged along lines that are parallel to the calorimeter axis, versus angular position of those lines,  $\phi$ , at time  $t = 25$  min. Again the temperature profiles for Tests 1 and 2 are similar and fairly uniform at roughly  $700^{\circ}\text{C}$ . In Test 3 the top of the calorimeter is less engulfed (Fig. 4c) and much cooler than it was in Tests 1 and 2, but the bottom is hotter. The stronger

winds in Test 3 may allow more oxygen to enter the region below the calorimeter and mix with fuel vapor than entered and mixed during Tests 1 and 2. This may be the reason for the elevated temperature near  $\phi = 270^{\circ}$  observed in Test 3.

Figure 6a shows the average fuel consumption rate versus average wind speed. This rate is the initial fuel mass divided by the fire duration (summarized in Table 1) and the pool surface area. The consumption rate increases nearly linearly with average wind speed. The increased fuel evaporation and entrainment may be caused by higher convection and radiation heat transfer to the fuel surface.

The average calorimeter temperature rise for each test (Table 1) is plotted against wind speed in Fig. 6b. The average temperature rise at the end of each test ( $t = D$ ) decreased nearly linearly with wind speed. This was caused by at least two effects. One was that the fire duration decreased as the wind speed increased. The second was that the calorimeter was less engulfed at higher wind speed than it was at lower speeds. The line marked  $t = 25$  min eliminates the fire duration effect. The rate at which energy from the fire was transferred to the calorimeter was not strongly affected by light winds with  $S_A \leq 1.22$  m/s, but decreased at higher wind speed because the calorimeter was less engulfed.

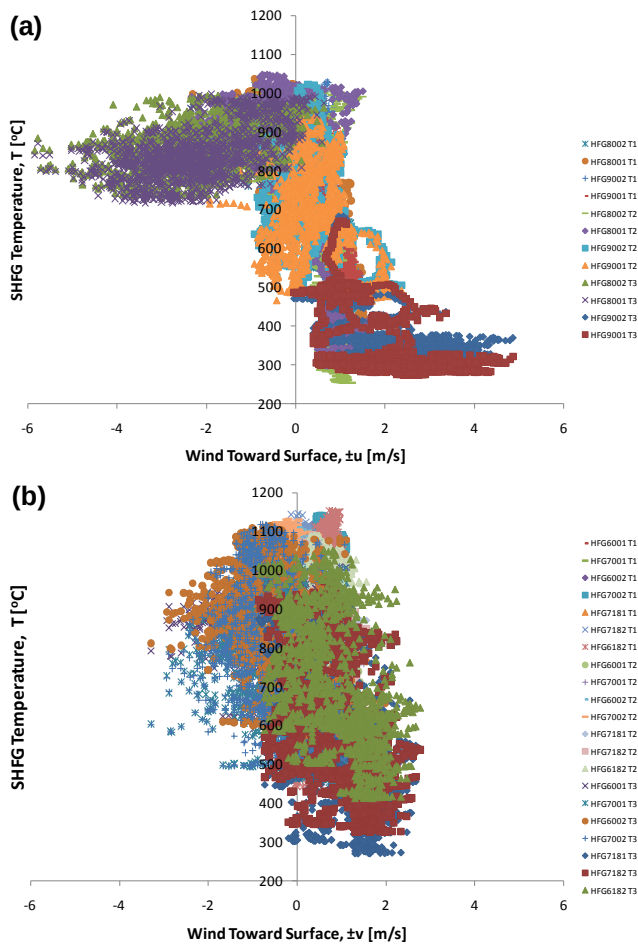


**Figure 6 (a) Fuel consumption rate versus average wind speed (b) Average calorimeter temperature rise at  $t = 25$  min and at the end of the fire ( $t = D$ ) versus average wind speed.**



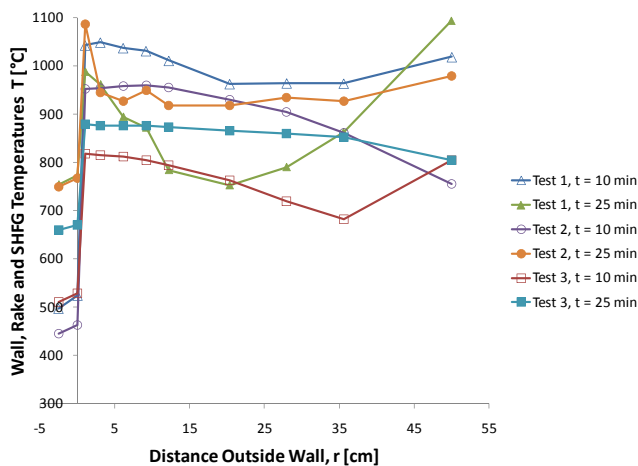
**Fire Environment** Heat transfer from the fire to the calorimeter is affected by the radiant flux, temperature, and speed of the participating media near the calorimeter surfaces. This section summarized measurements of these quantities so they may be used to develop and benchmark large-fire heat transfer models.

The SHFG temperatures characterize the radiant heat flux incident on their surfaces [Blanchat et al. 2002]. Figure 7a shows the temperatures of gages 8001, 8002, 9001 and 9002, which are located near the east and west calorimeter lids. Those temperatures are plotted versus the wind speed *toward* the calorimeter wall nearest those gages. That is, the temperatures of gages 8001 and 8002, which are near the east lid, are plotted versus  $u$ , which is the velocity component toward the east lid (the average of the  $u$ -components measure by the three anemometers on the east pole are used). The temperatures of gages 9001 and 9002, which are near the west lid, are plotted versus  $-u$  (as measured by anemometers on the west pole). Data from Tests 1, 2 and 3 (T1, T2 and T3) are included. Similarly, Fig. 7b shows the temperatures of the heat flux gages on the north and south sides of the calorimeter versus the wind speed component toward those sides (SHFG 6181 did not function and its data are not included).



Tests 1 and 3 (results from Test 2 are not included because they are similar to those from Test 1). For Tests 1 and 2 the bottom probe speed is roughly constant at 5 m/s. The speed measured by the upper two probes is much more oscillatory and as much as three times larger. For Test 3 the speed measured by all three probes is much more oscillatory than it was for Tests 1 and 2. For all three tests, the speeds measured by the middle probe are greater than those measured by the other two.

Figure 9 shows the temperatures measured by the participating media thermocouple rake on the north side of the calorimeter (Fig. 1b) versus distance from the calorimeter exterior surface,  $r$ . Results are presented for  $t = 10$  min (open symbols) and 25 min (solid symbols), and for all three tests (different symbol shapes are used for each test). The measured temperatures oscillate rapidly with time, so 11-second window averages are used. As mentioned earlier, these thermocouples hang beneath the SHFG fixture at the angular position  $\phi = 0$ , between thermocouple Rings 2 and 3. The average of the temperatures measured at  $\phi = 0$  of Rings 2 and 3 is plotted at  $r = -2.54$  cm (the position of the inner surface). The exterior surface temperature, as determined from the inner surface temperature using the Sandia One Dimensional Direct and Indirect Thermal (SODDIT) code [Blackwell et al. 1987] is plotted at  $r = 0$ . The average temperature of the four SHFGs whose fixture the thermocouples are hanging from is plotted at  $r = 50$  cm. This location was chosen somewhat arbitrarily, but is intended to indicate the media temperature “far” from the wall.



**Figure 9 Participating media thermocouple rake, wall and SHFG temperatures versus distance from the calorimeter outer surface. Data are for times  $t = 10$  and 15 minutes for Tests 1, 2 and 3.**

The participating media temperature at  $r = 1$  cm is significantly hotter than the wall. The media temperature at that location is actually close to that measured by the heat flux gages. The media temperature decreases fairly gradually with increasing distance from the wall. However, this spatial gradient is steeper at  $t = 25$  min in the low wind tests (Tests 1 and 2) than it is for the other conditions. Modeling of heat transfer in the participating media is required to understand this difference. This will be done in future work.

## V. SUMMARY

A 2.44-m-diameter, 4.57-m-long, 2.54-cm-wall thickness pipe calorimeter was suspended 1-m above a 7.93-m-diameter pool with 7.57 m<sup>3</sup> of jet fuel. The wind conditions and calorimeter temperature were measured at several locations as functions of time in three fire tests. The radiant heat flux, speed and temperature of the far participating media were measured near the calorimeter surface.

Tests 1 and 2 had relatively light winds and lasted roughly 40 minutes. Test 3 had much stronger winds and consumed the fuel in 25 minutes. The stronger winds in Test 3 tilted the flame cone so that the calorimeter was much less engulfed than in Tests 1 and 2. As a result the calorimeter temperature was hotter and more uniform in Tests 1 and 2 than it was in Test 3. The fire participating media radiant heat flux near each side of the calorimeter decreased as the wind speed component toward that surface increased. In all three tests, the participating media speed near the side of the calorimeter was greater near the elevation of the calorimeter midline than it was approximately 1 m above or below it. Those speeds were much more unsteady in the stronger wind test (Test 3) than it was in the other two. In all three fires, the temperature measure by a thermocouple 1 cm away from the side of the calorimeter was substantially higher than that of the calorimeter wall. Thermocouples further from the wall exhibit lower temperatures than the one 1 cm from the wall.

Fire models for transportation risk assessment studies must determine the fire duration (fuel consumption rate) and heat transfer from large fires to engulfed packages. Data from the experiments described in this paper can be used to develop and benchmark those models.

A website containing the full data set is available by contacting the first author. It may be used by other investigators to quantitatively benchmark large-fire simulations and models.

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